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MySQL 8.0 Command Line Cli  x  +  v
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE company_db;
Database changed
mysql> CREATE TABLE roles(
-> role_id INT PRIMARY KEY,
-> role_name VARCHAR(50)
-> );
ERROR 1050 (42501): Table 'roles' already exists
mysql>
mysql> CREATE TABLE users(
-> user_id INT PRIMARY KEY,
-> name VARCHAR(50),
-> email VARCHAR(50),
-> role_id INT
-> );
ERROR 1050 (42501): Table 'users' already exists
mysql>
mysql> CREATE TABLE departments(
-> dept_id INT PRIMARY KEY,
-> dept_name VARCHAR(50)
-> );
ERROR 1050 (42501): Table 'departments' already exists
mysql>
mysql> CREATE TABLE employees(
-> emp_id INT PRIMARY KEY,
-> emp_name VARCHAR(50),
-> dept_id INT,
-> manager_id INT,
-> salary DECIMAL(10,2)
-> );
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mysql>
mysql> CREATE TABLE customers(
-> customer_id INT PRIMARY KEY,
-> customer_name VARCHAR(50),
-> city VARCHAR(50)
-> );
ERROR 1050 (42501): Table 'customers' already exists
mysql>
mysql> CREATE TABLE orders(
-> order_id INT PRIMARY KEY,
-> customer_id INT,
-> order_date DATE,
-> total_amount DECIMAL(10,2)
-> );
ERROR 1050 (42501): Table 'orders' already exists
mysql>
mysql> CREATE TABLE accounts(
-> account_no INT PRIMARY KEY,
-> holder_name VARCHAR(50),
-> balance DECIMAL(10,2)
-> );
ERROR 1050 (42501): Table 'accounts' already exists
mysql>
mysql> CREATE TABLE faculty(
-> faculty_id INT PRIMARY KEY,
-> faculty_name VARCHAR(50)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE courses(
-> course_id INT PRIMARY KEY,
-> course_name VARCHAR(50),
-> faculty_id INT
-> );
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CREATE TABLE students(
-> student_id INT PRIMARY KEY,
-> student_name VARCHAR(50)
-> );
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Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE enrollments(
-> enrollment_id INT PRIMARY KEY,
-> student_id INT,
-> course_id INT,
-> );
Query OK, 0 rows affected (0.01 sec)

mysql> INSERT INTO roles VALUES
-> (1,'Admin'),
-> (2,'Manager'),
-> (3,'Employee'),
-> (4,'HR');
ERROR 1062 (23000): Duplicate entry '1' for key 'roles.PRIMARY'
mysql>
mysql> INSERT INTO users VALUES
-> (101,'Rahul','rahul@gmail.com',1),
-> (102,'Sneha','sneha@gmail.com',2),
-> (103,'Kiran','kiran@gmail.com',3);
ERROR 1062 (23000): Duplicate entry '101' for key 'users.PRIMARY'
mysql>
mysql> INSERT INTO departments VALUES
-> (1,'CSE'),
-> (2,'ECE'),
-> (3,'EEE'),
-> (4,'Mechanical');
ERROR 1062 (23000): Duplicate entry '1' for key 'departments.PRIMARY'
mysql>
mysql> INSERT INTO employees VALUES
-> (1,'Vishnu',1,NULL,80000),
-> (2,'Ravi',1,1,60000),
-> (3,'Kiran',2,1,45000),
-> (4,'Anil',3,2,55000),
-> (5,'Priya',1,2,70000);
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO customers VALUES
```

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mysql> INSERT INTO orders VALUES
-> (101,1,'2026-08-01',1200),
-> (102,2,'2026-08-02',3500),
-> (103,1,'2026-08-03',2800),
-> (104,3,'2026-08-04',5000);
Query OK, 4 rows affected (0.02 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO accounts VALUES
-> (101,'Raj',50000),
-> (102,'Ram',30000),
-> (103,'Sita',45000);
Query OK, 3 rows affected (0.02 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO faculty VALUES
-> (1,'Dr. Rao'),
-> (2,'Dr. Kumar');
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO courses VALUES
-> (101,'DBMS',1),
-> (102,'Java',2);
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO students VALUES
-> (1,'Amit'),
-> (2,'Neha');
Query OK, 2 rows affected (0.00 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO enrollments VALUES
-> (1,1,101),
-> (2,2,102);
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Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> SELECT emp_name,dept_name
-> FROM employees e
-> INNER JOIN departments d
-> ON e.dept_id=d.dept_id;
+-----+-----+
| emp_name | dept_name |
+-----+-----+
| Vishnu   | HR        |
| Ravi     | HR        |
| Kiran    | Finance   |
| Anil     | IT        |
| Priya    | HR        |
| Rahul    | HR        |
| Sneha    | Finance   |
| Arjun    | IT        |
| Priya    | IT        |
+-----+-----+
9 rows in set (0.00 sec)

mysql> SELECT name,email,role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id=r.role_id;
+-----+-----+-----+
| name  | email          | role_name |
+-----+-----+-----+
| Amit  | amit@gmail.com | Admin     |
| Neha  | neha@gmail.com | Manager   |
| Ravi  | ravi@gmail.com | Employee  |
| Priya | priya@gmail.com | NULL      |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT name,role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id=r.role_id;
+-----+-----+
| name | role_name |
+-----+-----+
|      |           |
|      |           |
|      |           |
|      |           |
+-----+-----+
3 rows in set (0.00 sec)
```

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3 rows in set (0.00 sec)

mysql> SELECT name,email,role_name
-> FROM users u
-> LEFT JOIN roles r
-> ON u.role_id=r.role_id
-> UNION
-> SELECT name,email,role_name
-> FROM users u
-> RIGHT JOIN roles r
-> ON u.role_id=r.role_id;
+-----+-----+-----+
| name  | email          | role_name |
+-----+-----+-----+
| Amit  | amit@gmail.com | Admin     |
| Neha  | neha@gmail.com | Manager   |
| Ravi  | ravi@gmail.com | Employee  |
| Priya | priya@gmail.com | NULL      |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT
-> e.emp_name AS Employee,
-> m.emp_name AS Manager
-> FROM employees e
-> LEFT JOIN employees m
-> ON e.manager_id=m.emp_id;
+-----+-----+
| Employee | Manager |
+-----+-----+
| Vishnu   | NULL    |
| Ravi     | Vishnu  |
| Kiran    | Vishnu  |
| Anil     | Ravi    |
| Priya    | Ravi    |
| Rahul    | NULL    |
| Sneha    | Rahul   |
| Arjun    | Rahul   |
| Priya    | Arjun   |
+-----+-----+
9 rows in set (0.00 sec)
```

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9 rows in set (0.01 sec)

mysql> SELECT
-> o.order_id,
-> customer_name,
-> city,
-> total_amount
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id=o.customer_id;

+-----+-----+-----+-----+
| order_id | customer_name | city | total_amount |
+-----+-----+-----+-----+
| 101 | Arjun | Hyderabad | 1200.00 |
| 102 | Meena | Vijayawada | 3500.00 |
| 103 | Arjun | Hyderabad | 2800.00 |
| 104 | Ramesh | Guntur | 5000.00 |
+-----+-----+-----+-----+

4 rows in set (0.00 sec)

mysql> SELECT
-> dept_name,
-> COUNT(*),
-> AVG(salary),
-> MAX(salary),
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id=d.dept_id
-> GROUP BY dept_name;

+-----+-----+-----+-----+
| dept_name | COUNT(*) | AVG(salary) | MAX(salary) |
+-----+-----+-----+-----+
| HR | 4 | 65000.000000 | 80000.00 |
| Finance | 2 | 52500.000000 | 60000.00 |
| IT | 3 | 63333.333333 | 70000.00 |
+-----+-----+-----+-----+

3 rows in set (0.00 sec)

mysql> SELECT
-> emp_name,
-> salary,
```

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-> salary,
-> dept_name
-> FROM employees e
-> JOIN departments d
-> ON e.dept_id=d.dept_id
-> WHERE salary>50000
-> AND dept_name='CSE';
Empty set (0.00 sec)

mysql> SELECT
-> customer_name,
-> SUM(total_amount) AS TotalPurchase
-> FROM customers c
-> JOIN orders o
-> ON c.customer_id=o.customer_id
-> GROUP BY customer_name
-> ORDER BY TotalPurchase DESC;

+-----+-----+
| customer_name | TotalPurchase |
+-----+-----+
| Ramesh | 5000.00 |
| Arjun | 4000.00 |
| Meena | 3500.00 |
+-----+-----+

3 rows in set (0.00 sec)

mysql> SELECT
-> student_name,
-> course_name,
-> faculty_name
-> FROM students s
-> JOIN enrollments e
-> ON s.student_id=e.student_id
-> JOIN courses c
-> ON e.course_id=c.course_id
-> JOIN faculty f
-> ON c.faculty_id=f.faculty_id;

+-----+-----+-----+
| student_name | course_name | faculty_name |
+-----+-----+-----+
| Amit | DBMS | Dr. Rao |
+-----+-----+-----+
```

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mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
-> SET balance=balance-3000
-> WHERE account_no=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE accounts
-> SET balance=balance+3000
-> WHERE account_no=999;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
-> SET balance=balance+2000
-> WHERE account_no=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT s1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
-> SET balance=balance-1000
-> WHERE account_no=101;
Query OK, 1 row affected (0.00 sec)
```

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mysql> SAVEPOINT s1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE accounts
-> SET balance=balance-1000
-> WHERE account_no=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO s1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM accounts;
+-----+-----+-----+
| account_no | holder_name | balance |
+-----+-----+-----+
| 101 | Raj | 47000.00 |
| 102 | Ram | 35000.00 |
| 103 | Sita | 45000.00 |
| 104 | Anil | 25000.00 |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```